

Δ vs frozen (good-oriented, value + 95% CI) · VANILLA-CONT-SSL block / SURGERY block · SCORE = mean per-metric-normalized (0-1) · BLUE BOX + big Δ = best arm per metric
WINNER row (champion duel — S=surgery V=vanilla-cont-SSL ==tie · tie = paired 95% CI overlaps 0): surgery 5 · vanilla-cont-SSL 4 · tie 4 → SURGERY

ViT-G vanilla continual-SSL	+8.55 [+6.14, +10.74]	+0.14 [+0.14, +0.15]	-0.01 [-0.02, +0.01]	+0.02 [+0.02, +0.03]	0 [0, 0]	+0.02 [+0.02, +0.03]	0 [0, 0]	+0.01 [+0.01, +0.01]	+0.01 [+0.01, +0.01]	-0.02 [-0.02, -0.01]	-0.02 [-0.03, -0.02]	-0.01 [-0.02, 0]	0 [-0.01, 0]	0.64
ViT-G surgery raw encoder	+6.41 [+3.95, +8.71]	+0.12 [+0.12, +0.13]	-0.01 [-0.02, +0.01]	+0.06 [+0.06, +0.06]	0 [0, 0]	+0.06 [+0.05, +0.06]	0 [0, 0]	0 [0, 0]	+0.02 [+0.02, +0.02]	-0.03 [-0.04, -0.02]	-0.01 [-0.02, -0.01]	-0.01 [-0.02, 0]	-0.04 [-0.04, -0.04]	0.57
ViT-G surgery 3stage_DI encoder	+4.88 [+2.41, +7.34]	+0.11 [+0.11, +0.11]	-0.01 [-0.02, +0.01]	+0.06 [+0.06, +0.06]	0 [0, 0]	+0.05 [+0.05, +0.05]	0 [0, 0]	0 [0, 0]	+0.02 [+0.01, +0.02]	-0.03 [-0.04, -0.02]	-0.01 [-0.01, 0]	-0.01 [-0.02, 0]	-0.02 [-0.03, -0.02]	0.60
ViT-G surgical autorgn encoder	+8.00 [+5.75, +10.25]	+0.18 [+0.17, +0.18]	-0.02 [-0.03, 0]	+0.03 [+0.03, +0.03]	0 [0, 0]	+0.03 [+0.03, +0.03]	0 [0, 0]	0 [0, 0]	+0.01 [+0.01, +0.01]	-0.02 [-0.03, -0.01]	-0.01 [-0.01, 0]	-0.02 [-0.03, -0.01]	-0.02 [-0.03, -0.02]	0.47
ViT-G surgery noDI encoder	+6.30 [+4.00, +8.60]	+0.12 [+0.11, +0.12]	-0.01 [-0.03, 0]	+0.06 [+0.06, +0.06]	0 [0, 0]	+0.05 [+0.05, +0.05]	0 [0, 0]	-0.01 [-0.01, -0.01]	+0.02 [+0.01, +0.02]	-0.02 [-0.03, -0.02]	-0.01 [-0.01, 0]	-0.03 [-0.04, -0.02]	-0.03 [-0.03, -0.03]	0.43
WINNER	=	S	=	S	V	S	V	V	S	=	S	=	V	
	action_top1 ↑ better classify motion class from frozen features	motion_cos ↑ better same-motion clips cluster vs different motions	taxonomy_f1 ↑ better read scene attributes from features	future_mse ↓ better predictor guesses masked next-frame tokens	rollout ↓ better error growth over multi-step rollout	causal ↓ better predict 2nd half from 1st half	tdist ↓ better accuracy decay vs prediction horizon	teacher_free ↓ better free-run minus teacher-forced gap	maskratio ↓ better degradation under heavier masking	aot ↑ better	tov ↑ better	pace ↑ better	tcc_tau ↑ better	SCORE (norm 0-1)